

Relative Exponential Periods on Polynomial Flags: Face Lattices, Stokes Generation, and Endpoint-Complete Relations

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Abstract

Let $k = \overline{\mathbb{Q}}$ and $k_{\mathbb{R}} = k \cap \mathbb{R}$. We study compact two-variable exponential integrals

$$F_{D,q,\Omega}(z) = \int_D e^{zq} \Omega$$

for polynomially parametrized real flag chains $D \subset \mathbb{R}^2$, real algebraic polynomial phases $q \in k_{\mathbb{R}}[x, y]$, and polynomial two-forms Ω . On the primitive observable Lefschetz locus we prove the geometric comparison theorem developed in the first part of the paper: vanishing of a common-phase function forces polynomial relative exactness, so the two-dimensional integral reduces, without denominators, to compact polynomial-line periods on the faces. We also identify the balanced face lattice with the toric quotient in relative homology, prove exact rooted Picard–Lefschetz observability, and realize irreducible weight-one intertwiners by relative Jacobian correspondences.

The second part closes the arithmetic endpoint problem on a stronger, explicitly testable sublocus, called the endpoint-complete Stokes–Lefschetz locus. The phase pushforward is retained as a compactly supported distribution in a completed relative Gauss–Manin module. Fourier–Laplace transformation converts multiplication by the exponential parameter into distributional differentiation. The interior singular behavior remains in the middle extension, while every newly supported endpoint or critical-value term is a punctual constituent and hence a compact polynomial-line boundary object. The Stokes multipliers of the primitive Fourier transform are read from the vanishing-cycle quiver. A strongly connected root quiver, together with the exponential torus, generates SL of the primitive solution space. Differential Galois theory then computes the complete affine orbit ideal of any finite family of flag solutions. Translation-active columns are free; split columns have no relations below full rank; and the only full-rank relations are the toric binomials of the joint determinant-character lattice.

The coefficient field is the fraction field of the actual compact polynomial-line boundary-function algebra. The complete polynomial-line theorem makes this boundary algebra normal and controls its specialization. Localizing at a nonsingular nonzero algebraic parameter gives a discrete valuation ring. The affine orbit quotient is flat there, so no new parameter-polynomial relation appears on endpoint evaluation. Beukers’ refined Siegel–Shidlovskii theorem therefore yields a complete numerical polynomial Kontsevich–Zagier theorem for endpoint-complete collections. For one endpoint-complete flag layer the primitive rank is either zero or at least two, while there is only one affine column; consequently the full-rank determinant case cannot occur. Thus every numerical polynomial relation for one layer is generated solely by algebraic pullback and trace, completed twisted Stokes, face and punctual relations, Jacobian correspondences when present, and the complete compact polynomial-line theorem.

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1 Introduction and main results

Kontsevich and Zagier proposed that numerical relations between periods are generated by additivity, algebraic change of variables, and Stokes’ formula [12]. In the exponential setting a differential form ω is replaced by $e^f\omega$, and the de Rham differential becomes $d_f = d + df\wedge$. Several formal, cohomological, and motivic definitions of exponential periods are compared in [4].

The one-dimensional compact polynomial-line category is unusually rigid. The companion paper [14] proves its linear Kontsevich–Zagier theorem, and [15] proves the complete polynomial theorem in that category. A direct extension to two variables is not automatic. Three new phenomena occur:

- (i) a sliced two-chain gives a relative curve cycle with moving endpoints;
- (ii) the affine fibre has a weight-one Jacobian part and boundary or infinity lattices;
- (iii) Beukers lifting produces polynomial coefficients in the exponential parameter, while relative antiderivative linearity concerns the outer integration variable.

The third point is a genuine logical distinction. It prevents a silent promotion of a relative functional theorem to a complete numerical polynomial theorem.

The purpose of this paper is to establish the geometric and differential algebraic comparison, and then to solve the arithmetic endpoint step on the explicit endpoint-complete Stokes–Lefschetz locus introduced below.

1.1 Polynomial flag periods

Put

$$k = \overline{\mathbb{Q}}, \quad k_{\mathbb{R}} = k \cap \mathbb{R}.$$

A polynomial flag chain is the pushforward of the standard square by a polynomial map

$$\Phi : [0, 1]^2 \longrightarrow \mathbb{R}^2, \quad \Phi \in k_{\mathbb{R}}[u, v]^2,$$

with the induced orientation and multiplicities. This includes ordinary triangles, rectangles, and graph flags

$$a \leq x \leq b, \quad A(x) \leq y \leq B(x), \quad A, B \in k_{\mathbb{R}}[x].$$

For $q, r \in k_{\mathbb{R}}[x, y]$, set

$$F_{D,q,r}(z) = \int_D r(x, y) e^{zq(x,y)} dx dy. \quad (1.1)$$

The phase is real on the real chain, so $q(D)$ is a compact interval. This one-dimensional phase support is essential for the Cauchy-transform separation used below.

Theorem 1.1 (Polynomial-flag E -functions). *Every function (1.1) is an E -function over k .*

The proof combines an elementary height estimate for the Taylor coefficients with holonomic direct image. The elementary estimate is included in Section 3.

1.2 Relative affine-Galois linearity

Let (K, D) be a differential field with algebraically closed constant field C . If $Y'_i \in K$, the Kolchin–Ostrowski theorem shows that every algebraic relation among the antiderivatives Y_i is generated by constant linear combinations already lying in K .

Theorem 1.2 (Relative affine-Galois linearity). *Let $Y_1, \dots, Y_m$ lie in a no-new-constants differential extension of K and satisfy $DY_i = f_i \in K$. After a matrix $S \in \mathrm{GL}_m(C)$, the transformed coordinates $Z = SY$ have the form*

$$Z_1 = g_1, \dots, Z_t = g_t \in K,$$

while $Z_{t+1}, \dots, Z_m$ are algebraically independent over K . Equivalently,

$$\ker(K[X_1, \dots, X_m] \rightarrow K[Y_1, \dots, Y_m])$$

is generated by affine-linear polynomials.

This theorem is exact for a variable outer endpoint. It does not control relations created after evaluating that endpoint. The distinction is made explicit in Remark 4.3.

1.3 Face lattices and observability

Let $\bar{f} : \bar{\mathcal{C}} \rightarrow S$ be a smooth proper family of connected curves, and let $B \subset \bar{\mathcal{C}}$ be a reduced divisor finite étale over S . Write

$$\mathcal{F}_B = \ker \left(\pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \right).$$

It is the balanced local system generated by the moving face points.

Theorem 1.3 (Face-lattice comparison). *There is a natural exact sequence*

$$0 \longrightarrow R_1 \bar{f}_* \mathbb{Z} \longrightarrow R_1 \bar{f}_*(\bar{\mathcal{C}}, B; \mathbb{Z}) \longrightarrow \mathcal{F}_B \longrightarrow 0.$$

For the generalized Jacobian $J_B = \text{Pic}^0(\bar{\mathcal{C}}/S, B)$ one has

$$1 \longrightarrow T_B \longrightarrow J_B \longrightarrow J \longrightarrow 0, \quad X^*(T_B) \simeq \mathcal{F}_B.$$

Thus the simplex face module is the lattice quotient in relative homology and, dually, the character lattice of the generalized-Jacobian torus. The two statements have opposite variance and should not be conflated.

The projective weight-one part is controlled by Picard–Lefschetz theory. If γ is a relative slice class and ν_i are the relative vanishing cycles, put an arrow $i \rightarrow j$ when the dual vanishing cycle of j pairs nontrivially with ν_i .

Theorem 1.4 (Exact rooted orbit theorem). *Let R_γ be the vertices reachable from those i for which the dual vanishing cycle pairs nontrivially with γ . Then*

$$\text{span}\{g\gamma - \gamma : g \in G\} = \text{span}\{\nu_i : i \in R_\gamma\}.$$

In particular, if the absolute vanishing cycles span the weight-one homology, their intersection graph is connected, and one incidence is nonzero, then the monodromy variation of γ is the complete weight-one homology.

1.4 The primitive observable Lefschetz locus

The denominator-free theorem requires the following hypotheses.

Definition 1.5 (Admissible observable flag datum). An admissible observable flag datum is a triple (D, q, Ω) with D a polynomial flag chain, $q \in k_{\mathbb{R}}[x, y]$, and $\Omega \in \Omega_{k[x, y]/k}^2$, satisfying:

- (A1) q has finite critical locus and every fibre $q^{-1}(t)$ is connected;
- (A2) the generic affine fibre has one place at infinity;
- (A3) after resolving the compactified pencil, every finite singular fibre has exactly one ordinary node and the finite critical values are distinct; over their complement the fibres form a smooth proper curve family, and the face intersections become a finite étale divisor after a finite base change;
- (A4) the resulting Lefschetz vanishing cycles span H_1 of the compactified generic fibre and have connected intersection graph;
- (A5) the sliced chain is nonzero on at least one regular phase interval, and on every interval on which it is nonzero it has nonzero incidence with at least one Lefschetz vanishing cycle.

The connected-fibre and finite-critical-locus assumptions put q in the primitive quasi-fibered class of Bonnet. The one-place-at-infinity condition removes an additional puncture lattice: it identifies the first homology of the affine generic fibre with that of its compactification.

Theorem 1.6 (Single-phase denominator-free functional theorem). *Let (D, q, Ω) be an admissible observable flag datum and put*

$$F(z) = \int_D e^{zq} \Omega.$$

If $F \equiv 0$, then there is a polynomial $a \in k[x, y]$ such that

$$\Omega = da \wedge dq.$$

Consequently

$$e^{zq} \Omega = d(e^{zq} a dq)$$

and

$$F(z) = \int_{\partial D} e^{zq} a dq.$$

The boundary relation is a compact polynomial-line functional relation and is therefore generated by the pullback–Stokes operations of [14].

The proof is in Section 7. Its algebraic step is Bonnet’s theorem that topological relative exactness equals polynomial relative exactness for primitive quasi-fibered polynomial maps [3].

1.5 Multiphase comparison

For a finite family, take a common punctured phase line S and let V_j be the weight-one local systems. The clean unconditional statement is the disjoint-block case.

Theorem 1.7 (Multiphase functional comparison: disjoint blocks). *Let (D_j, q_j, Ω_j) be admissible observable flag data. Assume that the complex local systems $V_{j, \mathbb{C}}$ are absolutely irreducible and pairwise nonisomorphic. If*

$$\sum_{j=1}^N c_j \int_{D_j} e^{zq_j} \Omega_j \equiv 0, \quad c_j \in k,$$

then the relation is generated by:

- (i) *linearity and algebraic pullback or trace;*
- (ii) *the polynomial identities $\Omega_j = da_j \wedge dq_j$ and phase-horizontal Stokes;*
- (iii) *balanced face relations and compact polynomial-line functional relations on the faces, including their zero-dimensional exponential boundary terms.*

For isomorphic blocks a relation can be diagonal across several copies. The flat intertwiner is nevertheless Hodge-theoretic.

Theorem 1.8 (Weight-one Hodge rigidity). *Let $\mathbb{V}$ and $\mathbb{W}$ be nonzero polarizable rational variations of effective weight one whose complex local systems are irreducible. Every nonzero flat map $\mathbb{V}_{\mathbb{C}} \rightarrow \mathbb{W}_{\mathbb{C}}$ is an isomorphism of complex variations of Hodge structure. After multiplication by a positive integer it is induced by a homomorphism of the relative Jacobians, and hence by an algebraic correspondence between the curve families.*

Hodge rigidity alone identifies the absolute weight-one systems; it does not automatically identify the relative slice normal functions. We therefore state the extension with an explicit compatibility condition in Section 9. This is one of the repairs made in passing from a research program to a theorem.

1.6 Endpoint-complete polynomial comparison

The functions in this paper are E -functions, so Beukers' refinement of the Siegel–Shidlovskii theorem lifts a polynomial relation at a nonsingular nonzero algebraic parameter to a polynomial-coefficient functional relation of the same degree. The issue is not the arithmetic lifting itself. The issue is to classify the resulting $k[z]$ -relation after the outer endpoint has been evaluated. Variable-endpoint Kolchin–Ostrowski theory does not do this: evaluation can create relations which were absent in the differential field of the outer variable.

We close this gap by retaining the phase pushforward before taking its Laplace transform. For a flag datum put

$$\langle \Theta_{D,q,\Omega}, \varphi \rangle = \int_D \varphi(q) \Omega.$$

It is a compactly supported distribution on the real phase line and

$$F_{D,q,\Omega}(z) = \langle \Theta_{D,q,\Omega}, e^{zt} \rangle.$$

The Weyl-algebra identities

$$\mathcal{L}(t\Theta) = \partial_z \mathcal{L}(\Theta), \quad \mathcal{L}(\partial_t \Theta) = -z \mathcal{L}(\Theta)$$

show exactly what polynomial coefficients in z do: they differentiate the phase distribution. Derivatives of a piecewise Gauss–Manin density have an interior Gauss–Manin part and distributions supported at critical or endpoint phase values. The latter are finite sums of $\delta_a^{(m)}$, whose Laplace transforms are $(-z)^m e^{az}$; they belong to the zero-dimensional part of the compact polynomial-line boundary category.

The primitive interior module is controlled by the Stokes group of its Fourier–Laplace transform. We use the following geometric locus.

Definition 1.9 (Endpoint-complete Stokes–Lefschetz collection). A finite collection $\mathcal{C}$ of polynomial flag data is *endpoint-complete Stokes–Lefschetz* when the following hold.

- (E1) Every datum is admissible observable in the sense of Definition 1.5.
- (E2) The primitive middle-extension Gauss–Manin module on the phase line is regular at infinity before Fourier transformation, its finite vanishing cycles form a basis of the full affine H_1 , and it has no additional weight-one constituent supported at infinity.
- (E3) In a stationary-phase basis, the nonzero rank-one Stokes arrows form a strongly connected directed graph, and the exponential torus separates their root characters.
- (E4) Let L_∂ be a differential normal closure of the actual compact polynomial-line boundary field over $k(z)$. For each primitive Picard–Vessiot field L_λ , the Galois quotient cut out by $L_\lambda \cap L_\partial$ factors through the central torus of the primitive group. Thus base change to the actual boundary field $K_\partial(\mathcal{C})$ defined below preserves the derived special-linear group. Distinct primitive simple blocks have no common projective quotient after this base change except those induced by algebraic Jacobian correspondences. Standard–dual identifications and rank-one twists are excluded unless an explicit algebraic correspondence supplies them.
- (E5) On every isotypic block the relative slice extension classes satisfy the correspondence compatibility of Definition 9.3; equivalently, all linear relations among the completed affine extension classes are generated by the stated correspondence, face, and phase-horizontal Stokes operations.

For one flag layer, (E5) follows from rooted observability: a nonzero primitive slice has nonzero variation, while a zero primitive slice is already reduced to the boundary by polynomial relative Stokes. The boundary-disjointness clause of (E4) remains to be checked.

Condition (E2) makes the completed Fourier transform a finite meromorphic connection whose exponential factors are indexed by the finite critical values. The topological Stokes formula of D'Agnolo–Hien–Morando–Sabbah identifies its off-diagonal multipliers with the vanishing-cycle quiver. Condition (E3) and the exponential torus then generate every elementary root subgroup, hence the full special linear group of each primitive block. Ramis density promotes this Stokes calculation to the differential Galois group.

Let $\mathcal{R}_\partial(\mathcal{C})$ be the finite functional algebra generated by all compact polynomial-line face functions, all punctual exponential terms required by the completed module, their differential closure, and z . The complete compact polynomial-line theorem of [15] gives, after its canonical linear reduction, a polynomial algebra over a finite exponential group algebra. In particular, $\mathcal{R}_\partial(\mathcal{C})$ is a normal noetherian domain. Put

$$K_\partial(\mathcal{C}) = \text{Frac } \mathcal{R}_\partial(\mathcal{C}).$$

This is the correct coefficient field: every endpoint forcing term is actually represented in the formal boundary category.

Theorem 1.10 (Endpoint-complete functional polynomial comparison). *Let $\mathcal{C}$ be endpoint-complete Stokes–Lefschetz. Over $K_\partial(\mathcal{C})$ the full polynomial relation ideal among the flag functions and a finite first-order differential closure is generated by:*

- (i) *the completed linear relations coming from algebraic pullback and trace, phase-horizontal polynomial Stokes, balanced faces, punctual endpoint terms, and relative Jacobian correspondences;*
- (ii) *the determinant-character binomials of the joint central quotient of the primitive differential Galois blocks.*

There are no other functional polynomial relations.

The determinant clause is explicit. After removing the translation-active columns and all constant column relations, a primitive block of rank n with fewer than n split columns has dense affine orbit and contributes no relation. A square full-rank block contributes its determinant coordinate. Across all square blocks, let $T_{\det} \subset \mathbb{G}_m^s$ be the image of the homogeneous differential Galois group under the determinant characters and put

$$\Lambda_{\det} = \{a \in \mathbb{Z}^s : \chi^a|_{T_{\det}} = 1\}.$$

The residual ideal is generated by

$$\delta^{a^+} - \eta_a \delta^{a^-}, \quad a \in \Lambda_{\det},$$

where $\eta_a \in K_\partial(\mathcal{C})^\times$. Clearing a boundary denominator writes each generator entirely in the formal compact boundary algebra.

Theorem 1.11 (Endpoint-complete numerical polynomial theorem). *Let $\mathcal{C}$ be endpoint-complete Stokes–Lefschetz and let $\xi \in k^\times$ be nonsingular for a common E -function system. Every polynomial relation over k among the values*

$$F_{D,q,\Omega}(\xi), \quad (D, q, \Omega) \in \mathcal{C},$$

is generated by specialization at $z = \xi$ of the operations in Theorem 1.10, multiplication and Fubini, and the complete compact polynomial-line theorem. Equivalently, the endpoint-complete formal period map defined in Definition 14.5 is injective.

The specialization step is local and denominator-free. The polynomial-line value theorem implies that the evaluation prime of the boundary algebra is exactly $(z - \xi)$. Its localization is therefore a discrete valuation ring. The affine orbit quotient is flat over that ring: linear and translation coordinates are polynomial, the determinant map is flat, and the residual lattice quotient is free because all partial-character coefficients are units. Hence a functional relation regular at ξ contracts to the local formal ideal. Beukers lifting supplies precisely such a functional relation with the prescribed numerical specialization.

Corollary 1.12 (One endpoint-complete flag layer). *For one endpoint-complete flag datum, every numerical polynomial relation at a nonsingular nonzero algebraic parameter is generated by completed twisted Stokes and the compact polynomial-line boundary theorem. No determinant binomial occurs.*

Indeed, a nonzero rational weight-one block has even rank $n \geq 2$, while one flag layer supplies only one affine solution column. After any splitting, its column rank is at most one and hence strictly smaller than n ; the special-linear orbit is dense. If the primitive block is zero, the original denominator-free relative Stokes theorem already reduces the period to the faces.

2 Polynomial flags and phase densities

2.1 Polynomial flag chains

Definition 2.1 (Polynomial two-chain). A polynomial two-chain over $k_{\mathbb{R}}$ is a finite k -linear combination of chains

$$D = \Phi_*[0, 1]^2, \quad \Phi \in k_{\mathbb{R}}[u, v]^2.$$

Its boundary is the alternating sum of the four polynomial face chains. A polynomial flag chain is a polynomial two-chain for which, after an affine change of the square coordinates,

$$\Phi(u, v) = (X(u), A(u) + vB(u))$$

with $X, A, B \in k_{\mathbb{R}}[u]$.

The chain language is preferable to set-theoretic domains: it records orientation, multiplicity, and algebraic change of variables without requiring injectivity of Φ .

For a polynomial two-form Ω , define

$$F_{D,q,\Omega}(z) = \int_D e^{zq} \Omega.$$

Pulling back to the square writes this as

$$F_{D,q,\Omega}(z) = \int_0^1 \int_0^1 P(u, v) e^{zQ(u,v)} du dv, \quad P, Q \in k[u, v]. \quad (2.1)$$

2.2 Gelfand–Leray density

Let $q \in k_{\mathbb{R}}[x, y]$ and let D be a real polynomial flag chain. Put $T = q(|D|) \subset \mathbb{R}$. Remove the critical values of q , the critical values of the restrictions of q to the faces, and the phase values of the face vertices. The complement is a finite union of open intervals e .

For $t \in e$, let

$$C_t = q^{-1}(t), \quad B_t = C_t \cap |\partial D|, \quad \gamma_t = D \cap C_t.$$

The last object is an oriented relative one-cycle in (C_t, B_t) , with the multiplicities inherited from D .

Let Ω be a polynomial two-form. On the smooth locus of q , choose a one-form α satisfying

$$dq \wedge \alpha = \Omega. \quad (2.2)$$

The restriction class of α to C_t is independent of the choice. Define

$$\rho_e(t) = \int_{\gamma_t} \alpha_t. \quad (2.3)$$

Lemma 2.2 (Phase-density formula). *After discarding chain components on which the pullback of Ω is zero, there is an L^1 density ρ on the compact set $T = q(|D|)$ such that*

$$F_{D,q,\Omega}(z) = \int_T e^{zt} \rho(t) dt. \quad (2.4)$$

On every regular interval e , the restriction ρ_e is real analytic and is a branch of a period of the algebraic Gauss–Manin system. In particular, it admits multivalued holomorphic continuation on the regular phase line. No claim that ρ_e is an algebraic function is made.

Proof. On a compact subinterval of e , the map q is a submersion on the sliced chain. The Gelfand–Leray or coarea formula gives

$$\int_{D \cap q^{-1}(e)} e^{zq} \Omega = \int_e e^{zt} \left(\int_{\gamma_t} \alpha_t \right) dt.$$

The inner integral is a period of the relative algebraic de Rham class $[\alpha_t]$ and therefore is real analytic on e and has the usual multivalued Gauss–Manin continuation. The coarea decomposition of the finite measure obtained by integrating the polynomial density on the compact chain shows that the resulting density is integrable across the finitely many critical and boundary values. Summing the regular intervals proves (2.4). A full-dimensional component on which q is constant cannot occur for nonconstant polynomial q ; a lower-dimensional component pulls every two-form back to zero. $\square$

2.3 Laplace uniqueness

Lemma 2.3 (Compact-support Laplace uniqueness). *Let ρ be a compactly supported integrable function on $\mathbb{R}$. If*

$$\int_{\mathbb{R}} e^{zt} \rho(t) dt = 0 \quad (z \in \mathbb{C}),$$

then $\rho = 0$ almost everywhere. If ρ is analytic on each member of a finite interval decomposition, it vanishes identically on every open interval.

Proof. Restriction to $z = is$, $s \in \mathbb{R}$, is the Fourier transform of the compactly supported L^1 function ρ . Fourier uniqueness gives almost-everywhere vanishing, and analyticity gives intervalwise vanishing. $\square$

For multiphase families it is useful to retain the Cauchy-transform form.

Lemma 2.4 (Cauchy jump separation). *Let $T \subset \mathbb{R}$ be a finite union of compact intervals and let ρ_j be piecewise analytic integrable densities on T . If*

$$\sum_j c_j \int_T e^{zt} \rho_j(t) dt = 0 \quad (z \in \mathbb{C}),$$

then

$$\sum_j c_j \rho_j(t) = 0$$

on every open regular interval.

Proof. All moments of the combined density vanish. Its Cauchy transform

$$\mathcal{C}(s) = \int_T \frac{\sum_j c_j \rho_j(t)}{s - t} dt$$

therefore vanishes for large s and hence on $\mathbb{C} \setminus T$. At an interior point of a regular interval, the two lateral values differ by $2\pi i$ times the analytic density. The jump is zero, so the density vanishes. $\square$

Remark 2.5 (Why real phase support is imposed). For a general complex-valued polynomial phase on a real two-chain, the pushforward is a two-dimensional measure in the complex phase plane. Its holomorphic moments can all vanish without the measure vanishing. Thus the one-variable Cauchy jump argument is unavailable. The phase-fibered hypothesis is mathematical content, not a notational convenience.

3 Polynomial-flag periods are E -functions

We use Beukers' normalization of Siegel E -functions. An entire function

$$F(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}$$

is an E -function if $a_n \in k$, the joint logarithmic height of the first n coefficients is $O(n)$, and F satisfies a nonzero linear differential equation over $k(z)$.

Theorem 3.1 (Polynomial-flag E -function theorem). *For $P, Q \in k[u, v]$, the function*

$$F(z) = \int_0^1 \int_0^1 P(u, v) e^{zQ(u, v)} du dv$$

is an E -function. Consequently every polynomial-flag period $F_{D, q, \Omega}$ is an E -function.

Proof. Termwise integration gives

$$F(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}, \quad a_n = \int_0^1 \int_0^1 P(u, v) Q(u, v)^n du dv.$$

Choose a number field K containing the coefficients and an integer $d \geq 1$ such that dP, dQ have algebraic-integer coefficients. The degree of PQ^n is $O(n)$, the number of its monomials is polynomial in n , and every archimedean conjugate of every coefficient is bounded by C^n for a fixed C . Writing

$$PQ^n = \sum_{i,j} c_{i,j,n} u^i v^j$$

gives

$$a_n = \sum_{i,j} \frac{c_{i,j,n}}{(i+1)(j+1)}.$$

If $N_n = O(n)$ bounds all occurring exponents, then

$$d^{n+1} \text{lcm}(1, \dots, N_n + 1)^2$$

clears all denominators. The prime number theorem in its elementary Chebyshev form gives $\log \text{lcm}(1, \dots, N) = O(N)$; hence the common denominator has logarithm $O(n)$. The conjugate and monomial-count estimates give the same bound for the joint height.

The integrand $P(u, v)e^{zQ(u, v)}$ is holonomic in (u, v, z) . Holonomic D -modules are stable under algebraic direct image, and integration over the fixed relative chain gives a holonomic function of z ; see [10, Chapters 1 and 3]. Therefore F satisfies a nonzero differential equation over $k(z)$. The integral is entire by uniform convergence on compact subsets. All three E -function conditions hold. $\square$

Theorem 3.2 (Beukers lifting). *Let $F_1, \dots, F_N$ be E -functions forming a vector solution of a first-order system over $k(z)$, and let $\xi \in k^\times$ be a nonsingular point. Every homogeneous polynomial relation of degree d among the values $F_i(\xi)$ lifts to a polynomial-coefficient functional relation, homogeneous of the same degree, specializing to the prescribed relation at $z = \xi$.*

Proof. This is the combination of [2, Theorems 1.3 and 3.2]; affine relations are homogenized by adjoining the constant E -function 1. $\square$

4 Relative affine-Galois linearity

Theorem 4.1 (Kolchin–Ostrowski). *Let K be a differential field of characteristic zero with algebraically closed constants C . Suppose $Y_1, \dots, Y_m$ lie in a no-new-constants differential extension and satisfy $DY_i \in K$. They are algebraically dependent over K if and only if some nonzero constant linear combination of them belongs to K .*

This is the classical antiderivative theorem; see [11, 18].

Let

$$R_Y = \{c \in C^m : c \cdot \mathbf{Y} \in K\}.$$

Theorem 4.2 (Affine-linear presentation). *Let $r = \dim_C R_Y$. There is $S \in \text{GL}_m(C)$ such that, for $\mathbf{Z} = S\mathbf{Y}$,*

$$Z_i = g_i \in K \quad (1 \leq i \leq r),$$

and $Z_{r+1}, \dots, Z_m$ are algebraically independent over K . Hence

$$\ker(K[\mathbf{X}] \rightarrow K[\mathbf{Y}]) = (Z_1 - g_1, \dots, Z_r - g_r).$$

Proof. Choose a basis of R_Y and extend it to a basis of C^m ; use these row vectors to form S . The first r transformed coordinates lie in K . If the remaining coordinates were algebraically dependent, Theorem 4.1 would put a nonzero constant linear combination of them in K , contradicting the choice of complement. Division by the monic linear polynomials $Z_i - g_i$ leaves a polynomial in algebraically independent variables, proving the kernel statement. $\square$

4.1 One iterated layer

Let x be an outer variable and let K be a differential field containing the inner period functions and their x -derivatives. If

$$V_i(x, z) = \int_{x_0}^x f_i(t, z) dt, \quad f_i \in K,$$

then Theorem 4.2 describes the complete polynomial relation ideal among the partial outer integrals over K .

Remark 4.3 (Endpoint evaluation is a separate problem). Take algebraically independent antiderivatives

$$V_i(x) = \int_b^x f_i(t) dt$$

over a differential field K . They have no functional algebraic relation over K , but $V_1(b) = \dots = V_m(b) = 0$. Endpoint evaluation therefore creates relations not present in the variable-endpoint function field. In particular, Theorem 4.2 cannot by itself justify a value-level polynomial theorem for completed two-variable periods.

5 The face lattice and the generalized Jacobian

Let $\bar{f} : \bar{\mathcal{C}} \rightarrow S$ be a smooth proper family of connected curves and let $\pi : B \rightarrow S$ be a nonempty reduced divisor finite étale over S . Define

$$\mathcal{F}_B = \ker \left(\pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \right). \quad (5.1)$$

Proposition 5.1 (Relative homology sequence). *There is a natural exact sequence of integral local systems*

$$0 \longrightarrow R_1 \bar{f}_* \mathbb{Z} \longrightarrow R_1 \bar{f}_*(\bar{\mathcal{C}}, B; \mathbb{Z}) \xrightarrow{\partial} \mathcal{F}_B \longrightarrow 0. \quad (5.2)$$

Proof. Apply the homology long exact sequence of the pair $(\bar{\mathcal{C}}_s, B_s)$. Connectedness gives $H_0(\bar{\mathcal{C}}_s, \mathbb{Z}) = \mathbb{Z}$, and the map $H_0(B_s, \mathbb{Z}) \rightarrow H_0(\bar{\mathcal{C}}_s, \mathbb{Z})$ is the degree map. The construction is locally constant in s and therefore yields (5.2). $\square$

Proposition 5.2 (Generalized-Jacobian torus). *Let $J = \text{Pic}^0(\bar{\mathcal{C}}/S)$ and $J_B = \text{Pic}^0(\bar{\mathcal{C}}/S, B)$. There is an exact sequence*

$$1 \longrightarrow T_B \longrightarrow J_B \longrightarrow J \longrightarrow 0, \quad T_B = \text{Res}_{B/S} \mathbb{G}_m / \mathbb{G}_m, \quad (5.3)$$

and

$$X^*(T_B) \simeq \mathcal{F}_B. \quad (5.4)$$

Proof. For reduced modulus, the generalized Jacobian is the extension of the ordinary Jacobian by the quotient of the Weil restriction of $\mathbb{G}_m$ by the diagonal $\mathbb{G}_m$; see [17, Chapter V]. Characters are contravariant. The character sequence associated to

$$1 \rightarrow \mathbb{G}_m \rightarrow \mathrm{Res}_{B/S} \mathbb{G}_m \rightarrow T_B \rightarrow 1$$

is

$$0 \rightarrow X^*(T_B) \rightarrow \pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \rightarrow 0,$$

which is (5.4). □

Remark 5.3 (Variance and weights). The relative-homology quotient $\mathcal{F}_B$ is a lattice of boundary classes. The same lattice is the character group of the torus T_B ; the torus itself is dual. In mixed-Hodge terms, relative homology and punctured-curve cohomology carry opposite Tate twists. The slogan “the face module is the toric part” is correct only after this variance is specified.

For a polynomial flag slice, B_t is the finite set of intersections with the oriented faces and

$$\partial\gamma_t \in \mathcal{F}_{B,t}.$$

The equation $\deg \partial\gamma_t = 0$ is simultaneously the balanced face relation and the boundary-of-boundary identity.

6 Rooted Picard–Lefschetz observability

Let H be a finite-dimensional vector space and suppose the monodromy group G is generated by relative Picard–Lefschetz transformations

$$(T_i - I)\xi = \varepsilon_i \langle \tilde{\nu}_i, \xi \rangle \nu_i, \quad \varepsilon_i \in \{\pm 1\}. \quad (6.1)$$

Here ν_i and $\tilde{\nu}_i$ are vanishing and dual vanishing cycles. We use the standard relative Picard–Lefschetz formula; see, for example, [9, Chapters 2–3]. Let γ be a relative slice class.

Definition 6.1 (Rooted variation graph). Put

$$I_\gamma = \{i : \langle \tilde{\nu}_i, \gamma \rangle \neq 0\}.$$

Put a directed edge $i \rightarrow j$ when

$$\langle \tilde{\nu}_j, \nu_i \rangle \neq 0.$$

Let R_γ be the vertices reachable from I_γ , and set

$$\mathcal{V}_\gamma = \mathrm{span}\{\nu_i : i \in R_\gamma\}, \quad \mathcal{O}_\gamma = \mathrm{span}\{g\gamma - \gamma : g \in G\}.$$

Theorem 6.2 (Exact rooted orbit theorem). *One has*

$$\mathcal{O}_\gamma = \mathcal{V}_\gamma.$$

The space is invariant under G .

Proof. The space $\mathcal{O}_\gamma$ is invariant because

$$h(g\gamma - \gamma) = (hg\gamma - \gamma) - (h\gamma - \gamma).$$

If $i \in I_\gamma$, (6.1) gives $\nu_i \in \mathcal{O}_\gamma$. If $\nu_i \in \mathcal{O}_\gamma$ and $i \rightarrow j$, then

$$(T_j - I)\nu_i = \varepsilon_j \langle \tilde{\nu}_j, \nu_i \rangle \nu_j,$$

so $\nu_j \in \mathcal{O}_\gamma$. Hence $\mathcal{V}_\gamma \subseteq \mathcal{O}_\gamma$. The same formula shows that every T_j preserves $\mathcal{V}_\gamma$.

Conversely, for every generator,

$$T_j\gamma - \gamma \in \mathcal{V}_\gamma.$$

If $g\gamma - \gamma \in \mathcal{V}_\gamma$, then

$$T_j g\gamma - \gamma = T_j(g\gamma - \gamma) + (T_j\gamma - \gamma) \in \mathcal{V}_\gamma.$$

The same argument applies to T_j^{-1} , since $T_j^{-1} - I$ has the same rank-one image. Induction on word length in the generators and their inverses gives $g\gamma - \gamma \in \mathcal{V}_\gamma$ for every g , proving the reverse inclusion. $\square$

On absolute curve homology the intersection graph is undirected.

Corollary 6.3 (Connected-graph observability). *Suppose the vanishing cycles span V , their intersection graph is connected, and $I_\gamma \neq \emptyset$. Then*

$$\mathcal{O}_\gamma = V.$$

Proof. Every vertex is reachable from the nonempty active set. Apply Theorem 6.2. $\square$

Proposition 6.4 (Absolute irreducibility). *Assume in addition that V carries a nondegenerate alternating pairing and that the T_i are the associated symplectic transvections. Under the hypotheses of Corollary 6.3, the representation on V is absolutely irreducible.*

Proof. Let $K/\mathbb{Q}$ be a characteristic-zero extension and let $0 \neq W \subset V_K$ be invariant. Choose $0 \neq w \in W$. Since the vanishing cycles span and the pairing is nondegenerate, some $\langle w, \nu_i \rangle$ is nonzero. Then $(T_i - I)w$ is a nonzero multiple of ν_i , so $\nu_i \in W$. Connectivity propagates membership to every vanishing cycle. Thus $W = V_K$. $\square$

7 Denominator-free relative exactness

7.1 Bonnet's theorem

For a polynomial map $q : \mathbb{C}^2 \rightarrow \mathbb{C}$, a polynomial one-form ω is *topologically relatively exact* if its restriction to every generic fibre has zero integral on every loop. It is *algebraically relatively exact* if

$$\omega = dR + a dq, \quad R, a \in \mathbb{C}[x, y].$$

Lemma 7.1 (Bonnet's hypotheses in relative dimension one). *Let $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ be nonconstant. If its critical locus is finite and every fibre is connected, then q is primitive and quasi-fibered in Bonnet's terminology.*

Proof. A finite critical locus has codimension two, so q is nonsingular in codimension one. Since the target has dimension one, Bonnet's requirement that the fibres be 1-generically connected means that every fibre is connected. His 2-generic nonemptiness condition is automatic: for every $c \in \mathbb{C}$, the proper ideal $(q - c) \subset \mathbb{C}[x, y]$ has a zero by the Nullstellensatz. The weaker generic connectedness and nonemptiness conditions defining primitivity follow as well. $\square$

Theorem 7.2 (Bonnet–Gavrilov relative exactness). *If $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ has finite critical locus and every fibre is connected, then every topologically relatively exact polynomial one-form is algebraically relatively exact.*

Proof. Apply Lemma 7.1 and [3, Theorem 1.5]. Bonnet also records the earlier two-variable connected-and-reduced theorem of Gavrilov. $\square$

7.2 From a zero phase density to relative exactness

Let Ω be a polynomial two-form. Choose a polynomial one-form ω with

$$d\omega = \Omega; \tag{7.1}$$

this is possible because $H_{\text{dR}}^2(\mathbb{A}^2) = 0$. On a regular fibre choose α as in (2.2). The Gelfand–Leray formula is

$$\nabla[\omega_t] = [\alpha_t]. \tag{7.2}$$

Lemma 7.3 (Observable density kills the Gelfand–Leray class). *Let (D, q, Ω) satisfy (A2)–(A5) of Definition 1.5. If the phase density vanishes on a regular interval on which the sliced chain is nonzero and has nonzero Lefschetz incidence, then*

$$[\alpha_t] = 0 \in H_{\text{dR}}^1(C_t)$$

for the generic affine fibre.

Proof. On the active interval in the hypothesis, the period $\int_{\gamma_t} \alpha_t$ vanishes and hence, by analytic continuation, on every continuation of that branch. Therefore $[\alpha_t]$ annihilates

$$\text{span}\{g\gamma_t - \gamma_t : g \in G\}.$$

By Corollary 6.3, this is the complete projective weight-one homology. The one-place-at-infinity hypothesis identifies this with $H_1(C_t)$ of the affine fibre. The de Rham–Betti pairing is perfect, so $[\alpha_t] = 0$. $\square$

Lemma 7.4 (The primitive one-form is fibrewise exact). *Under the hypotheses of Lemma 7.3, the restriction of ω to the generic fibre is exact.*

Proof. Equation (7.2) and Lemma 7.3 show that $[\omega_t]$ is horizontal. It is therefore monodromy invariant. For a Picard–Lefschetz transvection, invariance of the cohomology class forces its period on the corresponding vanishing cycle to be zero. Since the vanishing cycles span $H_1(C_t)$, all periods of ω_t vanish. Hence the restriction is exact. $\square$

Theorem 7.5 (Polynomial relative Stokes). *Let (D, q, Ω) be admissible observable. If the phase density vanishes on every regular interval, there is $a \in k[x, y]$ such that*

$$\Omega = da \wedge dq. \tag{7.3}$$

Consequently

$$e^{zq}\Omega = d(e^{zq}a \, dq). \tag{7.4}$$

Proof. By (A5) there is an active regular interval with nonzero Lefschetz incidence. The density vanishes there by hypothesis, so Lemmas 7.3 and 7.4 show that ω is topologically relatively exact. Apply Theorem 7.2:

$$\omega = dR + a dq$$

with $R, a \in \mathbb{C}[x, y]$. Taking exterior derivatives and using (7.1) gives (7.3). The coefficients descend to k : after fixing the finitely many monomials occurring in one solution, coefficient comparison is a finite linear system over k . Finally, $dq \wedge dq = 0$ gives (7.4). $\square$

Proof of Theorem 1.6. Functional vanishing and Lemma 2.3 make every regular phase density zero. Apply Theorem 7.5 and ordinary Stokes:

$$F(z) = \int_{\partial D} e^{zq} a dq.$$

Every face is polynomially parametrized, so the right side is a finite combination of compact polynomial-line functions. It vanishes identically because $F \equiv 0$. The functional formal theorem of [14] makes this boundary relation formal. $\square$

Remark 7.6 (Why one place at infinity appears). If the generic affine fibre has $r > 1$ points at infinity, then $H_1(C_t)$ contains an additional rank- $r-1$ puncture lattice. Projective Picard–Lefschetz observability does not detect it. One may replace (A2) by an explicit infinity-lattice observability hypothesis, but it cannot simply be omitted.

8 Hodge rigidity and Jacobian correspondences

Theorem 8.1 (Deligne uniqueness for irreducible factors). *An irreducible complex local system underlying a polarizable complex variation of Hodge structure carries such a variation uniquely up to tensoring by a rank-one complex Hodge structure $\mathbb{C}(r, s)$, equivalently up to a shift of the Hodge bigrading.*

This is [6, Proposition 1.13]; see also the convenient formulation in [1, Theorem 2].

Theorem 8.2 (Weight-one Hodge rigidity). *Let $\mathbb{V}$ and $\mathbb{W}$ be nonzero polarizable rational variations of Hodge structure of effective weight one on a connected smooth algebraic base. Assume their complex local systems are irreducible. Every nonzero flat map*

$$\phi : \mathbb{V}_{\mathbb{C}} \longrightarrow \mathbb{W}_{\mathbb{C}}$$

is a morphism of complex variations of Hodge structure.

Proof. Irreducibility makes ϕ an isomorphism of complex local systems. Transport the Hodge bigrading of $\mathbb{W}$ to $\mathbb{V}$. By Theorem 8.1, the transported grading differs from the original one by a shift $\mathbb{C}(r, s)$. Because both variations are effective of weight one and arise from rational Hodge structures, both types $(1, 0)$ and $(0, 1)$ occur. The shifted set of types can again be exactly $\{(1, 0), (0, 1)\}$ only when $r = s = 0$. Thus ϕ preserves the Hodge bigrading and hence the Hodge filtration. $\square$

Remark 8.3 (The reducible warning). The theorem fails for repeated reducible local systems. A constant elliptic curve gives a two-dimensional trivial local system; most constant matrices are flat but do not preserve $H^{1,0}$. The irreducibility hypothesis is therefore load-bearing.

Proposition 8.4 (Algebraic realization). *Let $f_i : \mathcal{C}_i \rightarrow S$ and $f_j : \mathcal{C}_j \rightarrow S$ be smooth proper curve families over a smooth connected complex algebraic base, with relative Jacobians J_i, J_j . A rational morphism of weight-one variations*

$$R^1 f_{i*} \mathbb{Q} \longrightarrow R^1 f_{j*} \mathbb{Q}$$

is induced, after multiplication by a positive integer and with the usual contravariance convention, by a homomorphism of abelian schemes between J_i and J_j . Its action on R^1 is represented by an algebraic divisor correspondence on $\mathcal{C}_i \times_S \mathcal{C}_j$.

Proof. The Hodge realization functor from abelian schemes over S to torsion-free polarizable integral variations of types $(-1, 0)$ and $(0, -1)$ is fully faithful; see [5, Rappel (4.4.3)]. After multiplying the rational morphism by an integer, full faithfulness therefore supplies the corresponding homomorphism of relative Jacobians.

The homomorphism and the Poincaré bundle determine a relative line bundle on $J_i \times_S J_j$. Pulling it back along the Abel–Jacobi maps (after an inessential finite base change if relative degree-one divisor classes are needed) gives a line bundle on $\mathcal{C}_i \times_S \mathcal{C}_j$. Its first Chern class, represented by a relative divisor with rational coefficients after rescaling, acts on R^1 as the original morphism. $\square$

Corollary 8.5 (Complex intertwiners are generated by correspondences). *Under the hypotheses of Theorem 8.2, every complex flat intertwiner is a complex linear combination of rational VHS morphisms and therefore, by Proposition 8.4, of algebraic Jacobian correspondences.*

Proof. For rational local systems, the intertwiner space is the kernel of a finite system of rational linear equations. Hence scalar extension gives

$$\mathrm{Hom}_{\pi_1}(\mathbb{V}, \mathbb{W}) \otimes_{\mathbb{Q}} \mathbb{C} \simeq \mathrm{Hom}_{\pi_1}(\mathbb{V}_{\mathbb{C}}, \mathbb{W}_{\mathbb{C}}).$$

Apply Theorem 8.2 and Proposition 8.4 to the rational summands. $\square$

9 Multiphase functional comparison

Let S be the phase line with all critical and boundary branch values removed. After a common finite étale base change, assume that all face points are sections. For each datum let

$$0 \rightarrow V_j \rightarrow H_j^{\mathrm{rel}} \rightarrow \mathcal{F}_j \rightarrow 0$$

be the relative-homology sequence and let γ_j be the sliced relative class. Let α_j denote the Gelfand–Leray cohomology class.

9.1 The direct-sum variation space

Set

$$V = \bigoplus_{j=1}^N V_j, \quad \Gamma = (\gamma_1, \dots, \gamma_N),$$

and

$$\mathcal{O}_{\Gamma} = \mathrm{span}\{g\Gamma - \Gamma : g \in \pi_1(S)\} \subseteq V. \tag{9.1}$$

The projection of $\mathcal{O}_{\Gamma}$ to V_j is $\mathcal{O}_{\gamma_j}$ and hence, for an observable datum, is all of V_j .

Lemma 9.1 (Disjoint irreducible factors). *Let $V_1, \dots, V_N$ be pairwise nonisomorphic irreducible complex representations of a group G . If a G -submodule $W \subseteq \bigoplus_j V_j$ projects nontrivially to every V_j , then*

$$W = \bigoplus_j V_j.$$

Proof. The finite-dimensional image algebra of $\mathbb{C}[G]$ is semisimple on the direct sum. A submodule is therefore a direct sum of some of the pairwise nonisomorphic simple factors. Nonzero projection to each factor forces every factor to occur. $\square$

9.2 Proof in the disjoint-block case

Theorem 9.2 (Disjoint-block functional comparison). *Under the hypotheses of Theorem 1.7, every constant-coefficient functional relation is generated by the operations listed there.*

Proof. By Lemma 2.4, the combined phase density vanishes on every open regular interval. Fix such an interval e and let J_e be the indices for which the sliced chain is nonzero on e . Analytic continuation to a base point of the common regular phase line gives, for every $g \in \pi_1(S)$,

$$\sum_{j \in J_e} c_j \langle \alpha_j, g \gamma_{j,e} \rangle = 0.$$

Subtract the equation for $g = 1$. The covector

$$\alpha_e = (c_j \alpha_j)_{j \in J_e} \in \left(\bigoplus_{j \in J_e} V_j \right)^\vee$$

annihilates the orbit-variation submodule associated with the tuple $(\gamma_{j,e})_{j \in J_e}$. By admissibility, its projection to every V_j , $j \in J_e$, is surjective. The factors remain pairwise nonisomorphic, so Lemma 9.1 makes the orbit submodule the whole direct sum. Hence $c_j \alpha_j = 0$ for every $j \in J_e$.

Every nonzero two-chain term is active on at least one regular interval. Repeating the argument over the finite interval decomposition therefore gives $c_j \alpha_j = 0$ for every index with $c_j \neq 0$. The proof of Theorem 7.5 then applies to the scaled form $c_j \Omega_j$ and gives a polynomial a_j with

$$c_j \Omega_j = da_j \wedge dq_j.$$

Stokes reduces the relation to the faces. The face terms are compact polynomial-line functional relations; they are generated by the formal operations of [14]. Their zero-dimensional endpoint terms are already part of the compact polynomial-line formal space. $\square$

9.3 Isomorphic blocks

Hodge rigidity makes every flat isomorphism between absolutely irreducible effective weight-one blocks Hodge-theoretic and hence algebraic on the relative Jacobians. A second condition is needed for the relative cycles.

Definition 9.3 (Relative correspondence compatibility). Fix an isotypic collection with common absolutely irreducible factor V , and use algebraic Jacobian correspondences to identify its copies of V . For a coefficient vector $m = (m_i)$, put

$$c_m(g) = \sum_i m_i (g \gamma_i - \gamma_i) \in V.$$

The collection is *relatively correspondence-compatible* if every m with $c_m = 0$ has the following geometric realization: after a finite base change and after applying a rational linear combination of algebraic correspondences to the γ_i , the relative cycle $\sum_i m_i \gamma_i$ differs from zero by a class in the balanced face lattice. The condition is imposed separately on every isotypic factor.

This condition is automatic when the slice cycles are literally identified by an algebraic change of variables, and when several amplitudes are placed on the same phase and the same slice. It is not a formal consequence of Hodge rigidity: a relative cycle is a normal-function datum, not only an absolute weight-one class.

Theorem 9.4 (Isotypic extension). *The conclusion of Theorem 9.2 remains true when pairwise nonisomorphism is replaced by relative correspondence compatibility.*

Proof. Decompose the direct sum into isotypic pieces $V_\lambda \otimes M_\lambda$. Absolute irreducibility gives $\text{End}_G(V_\lambda) = \mathbb{C}$, so every submodule is $V_\lambda \otimes M_\lambda^0$ for a subspace $M_\lambda^0 \subseteq M_\lambda$. By Corollary 8.5, every flat intertwiner between two copies is a complex linear combination of algebraic Jacobian correspondences. Relative correspondence compatibility permits the corresponding linear operations on the slice variation cocycles modulo the face lattices.

Choose a basis in which the orbit-variation submodule occupies the first multiplicity coordinates. The combined de Rham covector annihilates those coordinates, so their cohomology classes are fibrewise zero and Theorem 7.5 applies. The remaining transformed relative cycles have zero variation and, by the compatibility hypothesis, are represented modulo algebraic correspondences by face classes. The resulting relations are therefore generated by correspondences, phase-horizontal Stokes, and the boundary line theorem. $\square$

Remark 9.5 (The remaining normal-function issue). Without Definition 9.3, an invariant linear combination of relative slice classes can define a nontrivial admissible normal function even when the absolute local system has no fixed part. Hodge rigidity of the absolute H^1 does not eliminate this possibility. This is the precise remaining obstruction in the unrestricted isomorphic-block case.

10 A functional formal-period statement

Fix a finite collection $\mathcal{C}$ of admissible observable flag data. Let $\mathcal{P}_{\text{flag}}^{\text{fun}}(\mathcal{C})$ be the k -vector space generated by the symbols

$$[D, q, \Omega], \quad (D, q, \Omega) \in \mathcal{C},$$

together with the compact polynomial-line boundary symbols produced from the faces, modulo the following explicitly permitted relations:

- (F1) linearity in chains and differential forms;
- (F2) phase-preserving algebraic pullback and trace through auxiliary generically finite maps;
- (F3) the phase-horizontal Stokes identities

$$[D, q, da \wedge dq] = [\partial D, q, a dq], \quad a \in k[x, y];$$

- (F4) the compact polynomial-line functional relations, including their zero-dimensional exponential boundary symbols, of [14];

(F5) when $\mathcal{C}$ is relatively correspondence-compatible, the algebraic Jacobian-correspondence relations appearing in Definition 9.3.

The functional period map is

$$\text{per}_{\text{fun}, \mathcal{C}} : \mathcal{P}_{\text{flag}}^{\text{fun}}(\mathcal{C}) \longrightarrow \mathcal{O}(\mathbb{C}), \quad [D, q, \Omega] \longmapsto \left(z \mapsto \int_D e^{zq} \Omega \right).$$

Theorem 10.1 (Functional Kontsevich–Zagier theorem on the observable locus). *The map $\text{per}_{\text{fun}, \mathcal{C}}$ is injective if the weight-one blocks of $\mathcal{C}$ are pairwise nonisomorphic. It is also injective if $\mathcal{C}$ is relatively correspondence-compatible.*

Proof. This is Theorems 9.2 and 9.4, applied on each interval of the common regular phase decomposition and then summed. Every defining relation has zero functional period by algebraic change of variables, trace, correspondence functoriality, or Stokes, so the map is well defined. $\square$

11 Completed phase distributions and the boundary field

The functional theorem of Theorem 10.1 uses only the ordinary Laplace transform of the phase density. Polynomial coefficients in z force us to retain the action of the phase Weyl algebra. This section makes that completion precise and identifies every new boundary constituent with an object already covered by the compact polynomial-line theorem.

11.1 The compact phase distribution

Let (D, q, Ω) be a polynomial flag datum. Define a compactly supported distribution on the real phase line by

$$\langle \Theta_{D, q, \Omega}, \varphi \rangle = \int_D \varphi(q) \Omega, \quad \varphi \in C^\infty(\mathbb{R}). \quad (11.1)$$

Its support is contained in the compact interval $q(|D|)$. On the regular intervals of Section 2, it is represented by the analytic Gelfand–Leray density $\rho_e(t) dt$.

For a compactly supported distribution T , put

$$\mathcal{L}(T)(z) = \langle T(t), e^{zt} \rangle.$$

This is an entire function. With the convention $\langle \partial_t T, \varphi \rangle = -\langle T, \varphi' \rangle$, one has

$$\mathcal{L}(tT) = \partial_z \mathcal{L}(T), \quad \mathcal{L}(\partial_t T) = -z \mathcal{L}(T). \quad (11.2)$$

In particular,

$$F_{D, q, \Omega}(z) = \mathcal{L}(\Theta_{D, q, \Omega})(z). \quad (11.3)$$

Lemma 11.1 (Tensor and convolution compatibility). *If $T_1, \dots, T_m$ are compactly supported distributions on $\mathbb{R}$, then*

$$\prod_{j=1}^m \mathcal{L}(T_j) = \mathcal{L}(T_1 * \dots * T_m).$$

Moreover, the identities (11.2) commute with tensor product and convolution.

Proof. The external tensor product is compactly supported. Push it forward by the addition map $(t_1, \dots, t_m) \mapsto t_1 + \dots + t_m$. Evaluation on $e^{z(t_1 + \dots + t_m)}$ factors into the product of the evaluations. The Weyl identities follow from the distributional Leibniz rule and functoriality of pushforward. $\square$

The point of using distributions rather than repeatedly integrating by parts is that no endpoint term is discarded. The derivative of a piecewise Gauss–Manin density has three kinds of constituents: the middle-extension Gauss–Manin part on the punctured phase line, its singular germs at the critical values, and punctual subquotients supported at critical or boundary values. The punctual terms are completely elementary.

Lemma 11.2 (Punctual Fourier–Laplace terms). *A distribution supported at an algebraic point a is a finite sum $\sum_{m=0}^N c_m \delta_a^{(m)}$. Its Laplace transform is*

$$\sum_{m=0}^N c_m (-z)^m e^{az}.$$

Every such function lies in the differential algebra generated by the zero-dimensional exponential period e^{az} .

Proof. The structure theorem for point-supported distributions gives the first assertion. The second follows directly from $\langle \delta_a^{(m)}, e^{zt} \rangle = (-z)^m e^{az}$. $\square$

11.2 The completed relative Gauss–Manin module

Let $j : S \hookrightarrow \mathbb{A}_t^1$ be the complement of the critical values, face branch values, and endpoint phase values. Over S , the relative homology sequence of Proposition 5.1 and the Gelfand–Leray connection give a regular algebraic connection. Let $\mathcal{M}^{\text{mid}}$ be the direct sum of the middle extensions of its primitive weight-one constituents. Let $\mathcal{P}_a$ denote the finite regular holonomic punctual module generated by all local cohomology classes of the completed flag direct image at a .

Definition 11.3 (Completed phase module). *The completed phase module $\mathcal{M}_{D,q,\Omega}^{\text{cmp}}$ is the smallest regular holonomic $D_{\mathbb{A}_t^1}$ -subquotient of the algebraic direct image of the flag datum which contains the distribution class $\Theta_{D,q,\Omega}$, the primitive middle extension $\mathcal{M}^{\text{mid}}$, the balanced face module, and every punctual module $\mathcal{P}_a$ occurring at a critical or boundary phase value.*

The definition is intrinsic: it is the subquotient generated by the actual relative current, not an arbitrarily enlarged Picard–Vessiot system.

Proposition 11.4 (Finite completed closure). *For a polynomial flag datum, $\mathcal{M}_{D,q,\Omega}^{\text{cmp}}$ is regular holonomic and has finite rank away from finitely many phase values. Its Fourier–Laplace transform is holonomic. Under (E2), its nonpunctual primitive part is the Fourier transform of a regular middle-extension module which is regular at infinity, and every remaining composition factor is a balanced face factor or a punctual factor of Lemma 11.2.*

Proof. The exponential-free phase direct image is the algebraic direct image of a regular holonomic relative de Rham complex with a constructible relative chain condition. Regular holonomic modules are stable under algebraic direct image, subquotients, middle extension, and local cohomology. The chain has only finitely many critical, branch, and endpoint values, so only finitely many punctual factors occur. Fourier–Laplace transformation preserves holonomicity. Condition (E2) removes an additional weight-one constituent at infinity and identifies the nonpunctual primitive factor with the stated middle extension. See [10, 7] for the regular-holonomic and Fourier–Laplace formalism. $\square$

Corollary 11.5 (Endpoint completion is algebraic). *A finite differential closure of the flag functions can be chosen so that its inhomogeneous forcing coordinates are finite sums of compact polynomial-line face functions and functions $z^m e^{az}$ with $a \in k$. No analytic endpoint constant outside the compact polynomial-line category is introduced.*

Proof. Apply Fourier–Laplace transformation to Proposition 11.4. The face factors are direct images of polynomially parametrized one-chains. The punctual factors are described by Lemma 11.2. Choosing cyclic vectors in the finitely many holonomic factors produces a finite first-order closure with precisely these forcing terms. $\square$

11.3 The actual boundary-function algebra

Fix a finite collection $\mathcal{C}$. Let $\mathcal{R}_\partial(\mathcal{C})$ be the k -subalgebra of $\mathcal{O}(\mathbb{C})$ generated by z , by every compact polynomial-line face function occurring in the completed closures, by the finitely many exponentials e^{az} occurring in the punctual factors, and by the derivatives needed in the chosen first-order systems. We abbreviate it to $\mathcal{R}_\partial$ and put

$$K_\partial = \text{Frac}(\mathcal{R}_\partial).$$

Proposition 11.6 (Boundary normal form). *After quotienting by the complete linear compact polynomial-line relations and applying an invertible Laurent-polynomial change of the remaining moment coordinates, there are a finitely generated torsion-free additive group $\Gamma \subset k$ and algebraically independent coordinates $T_1, \dots, T_m$ such that*

$$\mathcal{R}_\partial \simeq k[z] \otimes_k k[\Gamma][T_1, \dots, T_m]. \quad (11.4)$$

Consequently $\mathcal{R}_\partial$ is a normal noetherian domain and its field of constants is k .

Proof. This is the finite functional normal form in the complete compact polynomial-line theorem [15]. The group Γ is a finitely generated subgroup of the additive group of a characteristic-zero field, hence is free abelian; its group algebra is a Laurent polynomial algebra. Formula (11.4) is therefore a Laurent polynomial algebra followed by an ordinary polynomial extension, so it is a normal noetherian domain. Every generator has Taylor coefficients in k . If a quotient f/g of two elements of this function field is a complex constant, comparison at the first nonzero Taylor coefficient of g shows that the constant lies in k . Thus the differential constant field is exactly k . $\square$

Proposition 11.7 (Boundary specialization prime). *Let $\xi \in k^\times$ be nonsingular for the finite boundary E -function system. Evaluation at $z = \xi$ has kernel*

$$\ker(\mathcal{R}_\partial \xrightarrow{\text{ev}_\xi} \mathbb{C}) = (z - \xi). \quad (11.5)$$

Hence

$$B_{\partial, \xi} := (\mathcal{R}_\partial)_{(z - \xi)} \quad (11.6)$$

is a discrete valuation ring with uniformizer $z - \xi$.

Proof. The quotient by $(z - \xi)$ is the formal compact polynomial-line value algebra at ξ . Its numerical period map is injective by the complete polynomial value theorem of [15]; therefore no further element vanishes at ξ . The ideal $(z - \xi)$ has height one in the normal noetherian domain of Proposition 11.6. The localization of a normal noetherian domain at a height-one prime is a discrete valuation ring. $\square$

12 Stokes root generation for the primitive blocks

We now compute the derived differential Galois group of the primitive Fourier–Laplace module. The argument uses only the rank-one Stokes arrows and a bounded graph condition.

12.1 Stationary-phase Stokes data

Let $\mathcal{M}$ be a primitive middle-extension module satisfying (E2). Write its distinct finite singular values as $c_1, \dots, c_n$. The vanishing-cycle-basis hypothesis identifies the stationary-phase solution space of $\widehat{\mathcal{M}}$ at infinity with

$$V = \bigoplus_{i=1}^n ke_i,$$

where the exponential factor on ke_i is $e^{c_i z}$. The exponential torus

$$T_{\text{exp}} \subset \text{GL}(V)$$

acts diagonally, and its character on the root line kE_{ij} is $\chi_i \chi_j^{-1}$.

The topological Fourier formula of D’Agnolo–Hien–Morando–Sabbah expresses the two Stokes multipliers in terms of the quiver of the original regular module. In the present rank-one vanishing-cycle situation, every nonzero off-diagonal quiver map supplies a nonzero root component E_{ij} in a Stokes logarithm; see [7, Theorem 5.4].

Definition 12.1 (Stokes root quiver). The Stokes root quiver Q_{St} has vertices $1, \dots, n$ and an arrow $i \rightarrow j$ whenever some Stokes logarithm has a nonzero E_{ji} -component. The indexing convention is immaterial; only the directed reachability relation is used.

Lemma 12.2 (Torus extraction of root lines). *Let $H \subset \text{GL}(V)$ be an algebraic group containing T_{exp} and a unipotent element u . Suppose the nonzero T_{exp} -weights occurring in $\log u$ are pairwise distinct. Then $\text{Lie}(H)$ contains every nonzero weight component of $\log u$.*

Proof. The adjoint orbit of $\log u$ under T_{exp} lies in $\text{Lie}(H)$. Its span is a direct sum of the weight lines occurring in $\log u$. Because the characters are distinct, evaluation at finitely many torus points gives a Vandermonde system which projects onto each weight component. Hence each component belongs to $\text{Lie}(H)$. $\square$

Lemma 12.3 (Strong root graphs generate special linear groups). *Let Q be a strongly connected directed graph on $\{1, \dots, n\}$. If a Lie subalgebra $\mathfrak{h} \subset \mathfrak{gl}_n(k)$ contains E_{ji} for every arrow $i \rightarrow j$ of Q , then*

$$\mathfrak{sl}_n(k) \subset \mathfrak{h}.$$

Consequently, an algebraic subgroup containing the corresponding root subgroups contains SL_n .

Proof. For a directed path $i = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_r = j$, repeated use of

$$[E_{i_{a+1}, i_a}, E_{i_{a+2}, i_{a+1}}] = -E_{i_{a+2}, i_a}$$

produces E_{ji} . Strong connectivity therefore gives every off-diagonal matrix unit. Then $[E_{ij}, E_{ji}] = E_{ii} - E_{jj}$ gives the trace-zero diagonal subalgebra. These matrices span $\mathfrak{sl}_n$. In characteristic zero, the connected algebraic subgroup with this Lie algebra contains the simply connected root group generated by the elementary unipotents, namely SL_n . $\square$

Theorem 12.4 (Stokes generation). *For a primitive endpoint-complete Stokes–Lefschetz block of rank n , the differential Galois group G of its Fourier–Laplace module over K_∂ satisfies*

$$\mathrm{SL}(V) \subset G. \quad (12.1)$$

Proof. Condition (E2) puts the primitive module in the regular-at-infinity setting of [7, Theorem 5.4]; its Fourier transform has the linear exponential factors $c_i z$. The Stokes formula identifies the nonzero root components with the arrows of Q_{St} . The exponential torus is part of the formal differential Galois group. Condition (E3) and Lemma 12.2 put every arrow root line in the Lie algebra of the analytic differential Galois group. Strong connectivity and Lemma 12.3 then give $\mathfrak{sl}(V)$. Ramis’ density theorem says that formal monodromy, the exponential torus, and the Stokes subgroups generate a Zariski-dense subgroup of the differential Galois group over $k(z)$; see [18, Chapter 8]. Thus that group contains $\mathrm{SL}(V)$. On extending the base to K_∂ , the new Galois group is the subgroup corresponding to the Picard–Vessiot intersection with the differential normal closure of K_∂ . Condition (E4) says that this quotient is central, so the derived subgroup $\mathrm{SL}(V)$ survives the base change. $\square$

12.2 Several primitive blocks

Decompose the semisimplification of the primitive homogeneous system as

$$M^{\mathrm{prim}} = \bigoplus_{\lambda \in \Lambda} V_\lambda \otimes E_\lambda,$$

where the V_λ are pairwise projectively nonisomorphic simple modules and E_λ are multiplicity spaces after the algebraic correspondence identifications of Section 8.

Proposition 12.5 (Product derived group). *Under (E4), the derived subgroup of the homogeneous differential Galois group contains*

$$\prod_{\lambda \in \Lambda} \mathrm{SL}(V_\lambda). \quad (12.2)$$

Any remaining coupling between the blocks factors through their determinant characters.

Proof. By Theorem 12.4, the projection to each simple block contains $\mathrm{SL}(V_\lambda)$. A proper subdirect product of simple special-linear factors can couple two factors only through an isomorphism of their adjoint simple quotients. Such an isomorphism identifies the corresponding projective differential modules, up to the standard–dual outer automorphism and a rank-one twist. Condition (E4) excludes these identifications unless they have already been combined by an explicit algebraic correspondence. The derived subgroup is therefore the product in (12.2). The quotient of each containing subgroup by its special-linear subgroup is measured by determinant, so every residual coupling is central and determinant-character valued. $\square$

Proposition 12.6 (Observable cyclic column). *Let $F(z)$ be the scalar flag function of one endpoint-complete datum and form a cyclic first-order system from F and its derivatives. Its projection to a primitive simple block is either zero or is one nonzero particular-solution column whose homogeneous differential module is the entire block. In the zero case, F belongs to the boundary field after the polynomial relative Stokes reduction.*

Proof. The differential submodule generated by the primitive projection of the cyclic vector is either zero or a nonzero submodule of a simple module; in the latter case it is the whole block.

Derivatives give the coordinates of that one cyclic column, not additional affine columns. If the primitive projection is zero, its Gelfand–Leray class annihilates the rooted monodromy orbit. By Theorem 6.2 and the perfect de Rham–Betti pairing it vanishes on the affine fibre. The proof of Theorem 7.5 then gives polynomial relative exactness and reduces F to its compact polynomial-line boundary terms. $\square$

13 Affine special-linear orbit ideals

We now pass from the homogeneous Stokes group to the complete polynomial relation ideal of the inhomogeneous flag solutions. The calculation is purely algebraic and is included in full.

Lemma 13.1 (Reductivity of the primitive homogeneous group). *Every primitive simple block occurring in an endpoint-complete collection has reductive differential Galois group in its faithful solution representation. After the correspondence identifications of (E4), the homogeneous group of the finite direct sum is reductive as well.*

Proof. Let G be the differential Galois group of a primitive simple block and let $U = R_u(G)$. The connected unipotent group U has a nonzero fixed vector by Lie–Kolchin. Because U is normal, its fixed space is G -stable; simplicity of the differential module makes that fixed space the full solution space. The faithful solution representation therefore kills U , so $U = 1$ and G is reductive. A finite product of these groups, enlarged only by the central determinant quotient allowed in (E4), is again reductive. $\square$

13.1 Affine Picard–Vessiot form

Let (K, D) be a differential field of characteristic zero with algebraically closed constant field k . Let

$$Y' = AY, \quad Y \in \mathrm{GL}_n(L),$$

be a fundamental matrix in a Picard–Vessiot field, and let

$$U' = AU + B, \quad U \in \mathrm{Mat}_{n \times r}(L), \quad B \in \mathrm{Mat}_{n \times r}(K). \quad (13.1)$$

In the endpoint-complete application, reductivity follows from Lemma 13.1. We state the affine calculation over an arbitrary reductive homogeneous group. For a differential automorphism σ one has

$$\sigma(Y) = Yg_\sigma, \quad \sigma(U) = U + Yc_\sigma.$$

Thus the affine differential Galois group is a subgroup of $\mathrm{Mat}_{n \times r} \rtimes G$.

Let W be its translation kernel. If G contains $\mathrm{SL}(V)$ on the standard module $V = k^n$, then

$$W = V \otimes R$$

for a uniquely determined subspace $R \subset k^r$: this follows from $\mathrm{End}_{\mathrm{SL}(V)}(V) = k$.

Lemma 13.2 (Translation reduction). *After a constant change of the r columns, suppose $R = \mathrm{span}(e_1, \dots, e_t)$. Then no polynomial relation over K involves the entries of the first t columns. Every remaining column is split: there are $A_0 \in \mathrm{Mat}_{n \times (r-t)}(K)$ and $C \in \mathrm{Mat}_{n \times (r-t)}(k)$ such that*

$$U_{\mathrm{sp}} = A_0 + YC. \quad (13.2)$$

Proof. The translation subgroup adds Yv independently to each of the first t columns. Since Y is invertible, these translations are Zariski dense in the corresponding affine coordinate spaces. A polynomial invariant under all of them is independent of those entries. For the remaining columns, projection of the affine translation term to $V \otimes (k^r/R)$ is a regular algebraic 1-cocycle of G . Since G is reductive in characteristic zero, it is linearly reductive and this cocycle is a coboundary. Thus there is a constant matrix C such that $U_{\text{sp}} + YC$ is fixed by the full affine Picard–Vessiot group. It therefore equals a matrix A_0 over K , which is (13.2) after replacing C by $-C$. $\square$

Constant column operations reduce C to a full-column-rank matrix and record all discarded columns as affine-linear relations over K . We now compute the remaining orbit.

Lemma 13.3 (Special-linear matrix orbits). *Let $C \in \text{Mat}_{n \times s}(k)$ have column rank $s \leq n$.*

- (i) *If $s < n$, the SL_n -orbit of C is the rank- s locus and is Zariski dense in $\text{Mat}_{n \times s}$.*
- (ii) *If $s = n$, the orbit is the hypersurface $\det X = \det C$.*

Proof. For $s < n$, any two injective maps $k^s \rightarrow k^n$ are carried to one another by an element of GL_n . The determinant of such an element can be corrected to one by scaling a complementary basis vector, without changing its action on the image of C . Hence SL_n is transitive on the rank- s locus. That locus is open dense. For $s = n$, left multiplication by SL_n preserves determinant, and if $\det X = \det C \neq 0$ then $XC^{-1} \in \text{SL}_n$. This proves the second assertion. $\square$

13.2 The joint determinant lattice

For several primitive blocks, perform Lemma 13.2 and the constant column reduction block by block. A block is *saturated* when the number of remaining independent split columns equals its rank. Only saturated blocks can contribute nonlinear relations. Enumerate them by $\lambda = 1, \dots, s$ and write

$$\Delta_\lambda = \det(U_\lambda - A_\lambda).$$

Let G^{hom} be the homogeneous differential Galois group and let

$$T_{\det} = \overline{\text{im}(G^{\text{hom}} \xrightarrow{(\det_1, \dots, \det_s)} \mathbb{G}_m^s)}.$$

Define its character lattice

$$\Lambda_{\det} = \{a \in \mathbb{Z}^s : x^a = 1 \text{ on } T_{\det}\}. \quad (13.3)$$

The lattice need not be saturated, because $T_{\det}$ may have a finite component group.

For $a \in \Lambda_{\det}$, the ratio

$$\eta_a = \frac{\Delta^{a^+}}{\Delta^{a^-}} \quad (13.4)$$

is fixed by the affine differential Galois group and hence belongs to $K^\times$.

Theorem 13.4 (Complete affine orbit ideal). *Assume the derived homogeneous group contains the product of the special linear groups of the pairwise projectively disjoint primitive blocks. Then the kernel of the evaluation map from the polynomial ring over K in all entries of U is generated by:*

- (i) *the affine-linear relations produced by the translation and constant column reductions;*
- (ii) *for the saturated blocks, the lattice binomials*

$$\Delta^{a^+} - \eta_a \Delta^{a^-}, \quad a \in \Lambda_{\det}. \quad (13.5)$$

There are no other polynomial relations.

Proof. Translation-active columns contribute a full affine-space factor by Lemma 13.2. Affine-linear column relations remove the constant kernel of each split matrix C . Every unsaturated block then has full column rank strictly smaller than its row rank, so Lemma 13.3(i) makes its orbit closure the full matrix space.

For a saturated block, Lemma 13.3(ii) says that the special linear derived group leaves only its determinant coordinate. After taking the product over all saturated blocks, the remaining orbit is the translate of the algebraic subgroup $T_{\det} \subset \mathbb{G}_m^s$ through the base determinant vector. The ideal of such a translated subtorus, including a possible finite component group, is the partial-character lattice ideal generated by (13.5); see [8]. Pulling this ideal back through the determinant maps and adjoining the already identified linear relations gives the full orbit-closure ideal. The Picard–Vessiot torsor identifies this orbit closure with the algebraic relation variety of the solution coordinates, proving completeness. $\square$

Corollary 13.5 (One-column freeness). *Let the primitive homogeneous rank be $n \geq 2$ and let U consist of one affine solution column which survives the affine-linear reduction of Lemma 13.2. Then the entries of that remaining column are algebraically independent over K . In particular, it contributes no nonlinear polynomial relation.*

Proof. If the translation kernel is nonzero, irreducibility makes it all of V , so translation gives a full affine-space orbit. If it is zero, the surviving split column has a nonzero constant column C , hence rank one. Since $1 < n$, Lemma 13.3(i) gives a dense orbit in k^n . Thus the remaining coordinate ring is a polynomial ring over K in either case. $\square$

13.3 Formal determinant-line reduction

Return to $K = K_{\partial}$. Every affine shift A_{λ} and every η_a in (13.5) belongs to the fraction field of the actual boundary algebra. Choose

$$\eta_a = b_a/c_a, \quad b_a, c_a \in \mathcal{R}_{\partial}, \quad c_a \neq 0.$$

Then (13.5) is equivalent to

$$c_a \Delta^{a^+} - b_a \Delta^{a^-} = 0. \tag{13.6}$$

Proposition 13.6 (Formal determinant-line realization). *Relation (13.6) is a formal tensor relation generated by exterior powers of the completed differential systems, the horizontal rank-one determinant-line identifications, multiplication/Fubini, and the compact polynomial-line boundary theorem. Different choices of b_a/c_a give the same formal relation.*

Proof. For a saturated block, exterior multiplication of its n split columns gives

$$\bigwedge^n (U_{\lambda} - A_{\lambda}) = (\det C_{\lambda}) \bigwedge^n Y_{\lambda}.$$

The right side is a solution of the rank-one determinant connection. A character $a \in \Lambda_{\det}$ makes the corresponding tensor product of determinant lines horizontal and trivial; its horizontal scalar is η_a . Expanding the exterior products gives the determinants Δ_{λ} , while representing the horizontal scalar in the boundary fraction field and clearing its denominator gives (13.6). Exterior expansion is an alternating sum of tensor products, hence is generated by multiplication and Fubini. The boundary coefficients are formal by [15]. If $b_a/c_a = b'_a/c'_a$, then $b_a c'_a - b'_a c_a = 0$ in $\mathcal{R}_{\partial}$; the complete boundary theorem makes the two cleared relations formally equal. $\square$

14 Local endpoint comparison and numerical specialization

The orbit theorem is over the boundary fraction field. To specialize at a fixed algebraic parameter we must prove that clearing denominators does not create a new relation at that parameter. The normal boundary model and a flatness argument give the required descent.

14.1 Regular local coefficients

Fix a nonsingular $\xi \in k^\times$ and put

$$B = B_{\partial, \xi}, \quad K = K_{\partial}, \quad \pi = z - \xi.$$

By Proposition 11.7, B is a discrete valuation ring.

Lemma 14.1 (Regularity of affine shifts). *The constant column reductions in Theorem 13.4 can be chosen so that every affine shift A_λ lies in B , entrywise. Every coefficient η_a in (13.4) is a unit of B .*

Proof. Choose a local fundamental matrix Y_λ holomorphic and invertible at ξ and use the actual entire E -function column U_λ . A split shift is $A_\lambda = U_\lambda - Y_\lambda C_\lambda$; it is holomorphic at ξ and belongs to K . Under the embedding of K into the meromorphic germs at ξ , the DVR valuation is the order at ξ , so holomorphy gives $A_\lambda \in B$. For a saturated block, both $\det Y_\lambda(\xi)$ and the selected nonzero constant minor $\det C_\lambda$ are nonzero. Hence every determinant ratio defining η_a has valuation zero. Thus $\eta_a \in B^\times$. $\square$

14.2 Flatness of the orbit model

Let $L = \Lambda_{\det} \subset \mathbb{Z}^s$ and let $\rho : L \rightarrow B^\times$ be $a \mapsto \eta_a$. Put

$$D_\rho = B[\delta_1, \dots, \delta_s]/I_\rho, \quad I_\rho = (\delta^{a^+} - \rho(a)\delta^{a^-} : a \in L). \quad (14.1)$$

Lemma 14.2 (Flat partial-character quotient). *The B -module D_ρ is free, hence flat.*

Proof. The lattice L defines a congruence on $\mathbb{N}^s$: two exponent vectors are equivalent when their difference lies in L . The character ρ identifies any two monomials in one congruence class by multiplication with a unit of B , and the character identity makes this identification independent of the chosen lattice path. Choose one monomial representative in each congruence class. Reduction along the binomials expresses every monomial as a unique unit multiple of its representative, and no nonzero element of B kills a representative. These representatives form a B -basis. Finite generation of the ideal follows from noetherianity, equivalently from the finite Markov basis theorem for lattice ideals [8]. $\square$

Lemma 14.3 (Flat determinant morphism). *For every $n \geq 1$, the homomorphism*

$$B[\delta] \longrightarrow B[x_{ij} : 1 \leq i, j \leq n], \quad \delta \longmapsto \det(x_{ij}),$$

is flat.

Proof. Flatness is local on source and target. The source local rings are regular, and the target local rings are regular and hence Cohen–Macaulay. Every geometric fibre is the hypersurface $\det X = c$ in affine n^2 -space; it has pure dimension $n^2 - 1$, including at $c = 0$. The dimension formula therefore holds at every local point. Miracle flatness [16, Theorem 23.1] gives the assertion. $\square$

Proposition 14.4 (Flat local orbit quotient). *Let I_B be the ideal over B generated by the regular affine-linear relations and the determinant lattice relations of Theorem 13.4. Then*

$$B[\mathbf{X}]/I_B$$

is B -flat. Moreover, if I_K denotes the complete generic relation ideal, then

$$I_K \cap B[\mathbf{X}] = I_B. \quad (14.2)$$

Proof. After constant row and column operations, the linear part of the orbit ideal is a coordinate ideal. Translation by the affine shifts $A_\lambda \in B$ is a polynomial automorphism and does not affect flatness. The remaining coordinate ring is obtained from D_ρ by base change along the product of the determinant morphisms and by adjoining the unsaturated matrix and translation coordinates. The determinant morphisms are flat by Lemma 14.3; D_ρ is B -flat by Lemma 14.2; polynomial extension preserves flatness. Thus the complete quotient is B -flat.

A flat module over the DVR B is torsion-free and injects into its generic fibre. Therefore a polynomial in $B[\mathbf{X}]$ which lies in I_K already vanishes in the B -quotient, proving (14.2). $\square$

14.3 The endpoint-complete formal algebra

Let $\mathcal{C}$ be endpoint-complete and fix ξ as above. Start with the commutative k -algebra generated by the flag symbols of $\mathcal{C}$, their finite differential closures, and all boundary symbols in $\mathcal{R}_\partial$. Localize only at boundary elements whose value at ξ is nonzero.

Definition 14.5 (Local endpoint-complete formal algebra). The algebra $\mathcal{P}_{\mathcal{C},\xi}^{\text{ec}}$ is the preceding localized symbol algebra modulo the following relations:

- (K1) linearity in chains and forms, and algebraic pullback and trace;
- (K2) multiplication, Cartesian product/Fubini, and the zero-dimensional exponential product and phase-shift identities;
- (K3) the Weyl identities (11.2) in the completed phase module and all differential consequences of its algebraic first-order system;
- (K4) phase-horizontal polynomial Stokes, balanced face relations, punctual endpoint relations, and the complete compact polynomial-line formal relations;
- (K5) the algebraic Jacobian-correspondence and relative extension relations supplied by (E4)–(E5);
- (K6) the cleared determinant-line relations (13.6) for a finite Markov basis of Λ_{det} .

Its functional period map takes a symbol to the germ at ξ of the corresponding entire function. Its numerical period map evaluates that germ at ξ .

Every listed operation is an identity of algebraic exponential periods. The localization is harmless: its denominators are boundary functions known to be nonzero at the chosen parameter.

Theorem 14.6 (Local parameter-polynomial formal comparison). *For an endpoint-complete Stokes–Lefschetz collection, the kernel of the functional period map on $\mathcal{P}_{\mathcal{C},\xi}^{\text{ec}}$ is zero. Equivalently, every polynomial-coefficient functional relation in a finite E -function closure, with coefficients regular at ξ , is generated by (K1)–(K6) over $B_{\partial,\xi}$.*

Proof. By Corollary 11.5, all nonprimitive forcing terms belong to the boundary algebra. Conditions (E2)–(E4) and Theorem 12.4 and Proposition 12.5 give the product of special-linear derived groups. Condition (E5) identifies the complete affine-linear reduction with the formal relations in (K1), (K3)–(K5). The generic relation ideal is therefore exactly the ideal of Theorem 13.4. Its nonlinear generators are formal by Proposition 13.6 and (K6).

All coefficients occurring in a relation regular at ξ lie in the DVR $B = B_{\partial, \xi}$. By Lemma 14.1 the displayed orbit generators also lie over B , and by Proposition 14.4 the generic ideal contracts to their B -ideal. Thus no additional local functional relation exists. Finally, the boundary normal form of Proposition 11.6 identifies the symbolic boundary algebra with its actual function algebra, so the descent is faithful. $\square$

Theorem 14.7 (Numerical endpoint-complete polynomial comparison). *Let $\xi \in k^\times$ be nonsingular. The numerical period map*

$$\text{per}_\xi : \mathcal{P}_{\mathcal{C}, \xi}^{\text{ec}} \longrightarrow \mathbb{C}$$

is injective. Hence every polynomial relation over k among the flag values at ξ is generated by (K1)–(K6).

Proof. First clear the finitely many localized boundary denominators; their values at ξ are nonzero by construction. Choose a finite first-order E -function system containing the flag functions, the boundary functions, and the closure coordinates. Beukers’ refinement of Siegel–Shidlovskii lifts a prescribed polynomial relation among the values at ξ to a polynomial-coefficient functional relation of the same degree whose specialization is the prescribed polynomial; see Theorem 3.2. The lifted relation is regular at ξ . By Theorem 14.6 it is a combination of the local formal generators. Specializing this formal combination at $z = \xi$ gives the original numerical relation. Boundary coefficients specialize faithfully by Proposition 11.7. Therefore the numerical kernel is zero. $\square$

Corollary 14.8 (Complete one-layer theorem). *Let (D, q, Ω) be one endpoint-complete flag datum and let $\xi \in k^\times$ be nonsingular. Every polynomial relation over k among $F_{D, q, \Omega}(\xi)$, its algebraically required closure coordinates, and the associated compact boundary values is generated by (K1)–(K5). After eliminating the closure coordinates, every relation in the single value $F_{D, q, \Omega}(\xi)$ over the boundary value algebra is formal. No determinant-line generator is needed.*

Proof. If the primitive projection is zero, Proposition 12.6 reduces the function to the boundary algebra. Otherwise its rational weight-one rank is even and at least two. The cyclic first-order closure supplies one affine particular-solution column, and its nonzero primitive projection means that this column survives the affine-linear reduction. By Corollary 13.5, one column cannot saturate a rank- n block, so $\Lambda_{\det}$ contributes no relation. Apply Theorem 14.7 and restrict its injective period map to the subalgebra generated by the original flag value and the boundary values. $\square$

15 Examples and falsification checks

15.1 A triangular elliptic endpoint-complete example

Consider

$$q(x, y) = y^2 - x^3 + 3x, \quad E_t : y^2 = x^3 - 3x + t.$$

The only affine critical points are $(1, 0)$ and $(-1, 0)$, with critical values 2 and -2 ; both singular fibres are nodal. The projective pencil is

$$Y^2 Z - X^3 + 3X Z^2 - t Z^3 = 0.$$

Its unique point at infinity is $p_\infty = [0 : 1 : 0]$, and the derivative with respect to Z at p_∞ is $Y^2 = 1$. Hence the total pencil is smooth near infinity and there is no vanishing cycle at infinity. The two finite vanishing cycles form a basis of the rank-two affine H_1 .

At $t = 0$ the branch points of the x -projection are $-\sqrt{3}, 0, \sqrt{3}$. The lifts of the adjacent intervals $[-\sqrt{3}, 0]$ and $[0, \sqrt{3}]$ are vanishing cycles with intersection ± 1 . The two opposite Stokes sectors in the Fourier transform therefore supply the two root arrows of $\mathfrak{sl}_2$ by the formula of [7, Theorem 5.4]. Since the exponential factors e^{2z} and e^{-2z} are distinct, the exponential torus separates them. Thus (E2)–(E3) hold.

The vertical face has a quadratic restricted phase and hence a rank-one homogeneous boundary module. The two slanted faces have the same restricted cubic phase

$$-u^3 + \frac{5}{2}u^2 + \frac{1}{4}u - \frac{3}{8},$$

whose critical values differ by $28\sqrt{7}/27$, not by 4, the difference of the primitive exponential factors 2 and -2 . Over the fixed exponential parameter z , a projective identification can twist all exponential factors by one common rank-one character but cannot dilate z or change their pairwise differences. Every exponential difference in a tensor construction of the cubic boundary block lies in the lattice $(28\sqrt{7}/27)\mathbb{Z}$, whereas $4 \notin (28\sqrt{7}/27)\mathbb{Z}$. Thus even the adjoint tensor categories have no common projective simple quotient with the elliptic block, and (E4) holds.

Take the rational triangle

$$D = \text{conv} \left\{ \left(-\frac{1}{2}, -1 \right), \left(-\frac{1}{2}, 1 \right), \left(\frac{1}{2}, 0 \right) \right\}.$$

The slice $E_0(\mathbb{R}) \cap D$ is a relative arc through $(0, 0)$. Near that point,

$$x = -\frac{y^2}{3} + O(y^6).$$

The simplex arc is tangent to the real y -axis, while the adjacent vanishing cycle is tangent to the imaginary y -axis. Their incidence is ± 1 , so Corollary 6.3 gives observability. For a single layer (E5) is automatic. Consequently this is an endpoint-complete one-layer example, and Corollary 14.8 applies to every polynomial amplitude.

For the special amplitude $2y \, dx \wedge dy$, the reduction can be seen directly. Taking $a = x$ gives

$$da \wedge dq = 2y \, dx \wedge dy, \quad 2ye^{zq} \, dx \wedge dy = d(xe^{zq} dq).$$

This checks that the completed theorem agrees with ordinary polynomial relative Stokes when the primitive class is zero.

15.2 The affine-rank threshold

The smallest special-linear orbit calculations test the nonlinear part of the proof. For $n = 2$ and one split column, SL_2 is transitive on $\mathbb{A}^2 \setminus \{0\}$, whose closure is all of $\mathbb{A}^2$; hence there is no polynomial relation. With two independent split columns, the orbit is

$$\{X \in \text{Mat}_{2 \times 2} : \det X = \det C\}.$$

Thus the first possible nonlinear relation occurs exactly when the number of independent split columns reaches the primitive rank. If the determinant image contains $\mathbb{G}_m$, even this equation disappears after taking the orbit closure. These two cases verify the dichotomy in Lemma 13.3 and Theorem 13.4.

15.3 Why connected fibres are necessary

Let

$$q = x^2, \quad \omega = x \, dy.$$

Every regular fibre is the disjoint union of the two lines $x = \pm\sqrt{t}$, and ω is exact on each component. If $\omega = dR + a \, dq$ with polynomials R, a , comparison of dy coefficients gives $R = xy + h(x)$; comparison of dx coefficients would require $y + h'(x) + 2xa = 0$, impossible polynomially. Thus topological relative exactness need not be denominator-free when the fibres are disconnected.

15.4 Why isomorphic absolute blocks are not enough

Two identical weight-one local systems can carry different relative slice normal functions. Hodge rigidity identifies the absolute systems, but a linear combination of the relative classes can be monodromy invariant without being a face class. This is the normal-function obstruction recorded in Remark 9.5 and is why relative correspondence compatibility is retained in (E5). The endpoint theorem removes specialization relations; it does not manufacture a missing geometric identification of extension classes.

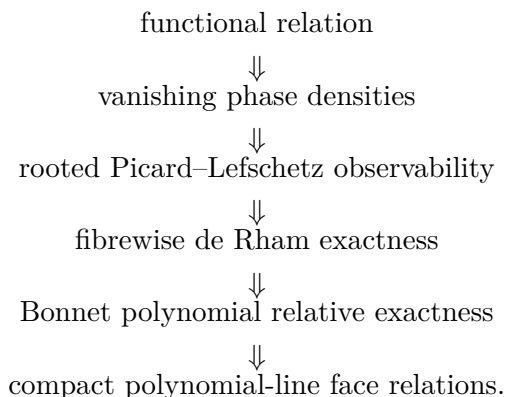
15.5 Why outer affine linearity alone remains insufficient

Let Y_1, Y_2 be algebraically independent antiderivatives normalized by $Y_1(b) = Y_2(b) = 0$. Their variable-endpoint relation ideal is zero, but their endpoint values satisfy two new linear equations. Thus Theorem 4.2 alone cannot imply a value theorem. The repair in Sections 11 to 14 uses additional information absent from this example: the punctual completion, the primitive Stokes group, the actual boundary function field, and flat local descent at the chosen parameter.

16 Scope and remaining boundaries

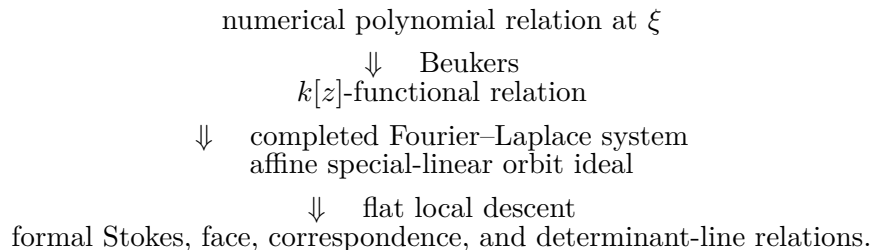
There are two unconditional levels in the paper.

On the primitive observable Lefschetz locus, without the stronger endpoint hypotheses, the paper proves the constant-coefficient functional chain



On the endpoint-complete Stokes–Lefschetz locus, the completed phase module and Stokes

calculation strengthen this to the full value-level polynomial chain



For one endpoint-complete layer the determinant line never occurs, so the terminal formal relations are exactly the completed linear operations and the compact polynomial-line theorem.

The endpoint-complete hypotheses are stronger than admissible observability and are not asserted to hold for every polynomial phase. The remaining geometric boundaries are the following.

First, several places at infinity produce a weight-one puncture or 1-motive constituent. It can in principle be added to the completed module, but its Stokes root quiver and affine extension classes require a separate observability theorem.

Second, non-tame or resonant Fourier transforms can have repeated exponential factors and larger Stokes blocks. The root-extraction proof must then be replaced by a block-level Stokes generation theorem; strong connectivity of the rank-one quiver is no longer the correct condition.

Third, unrestricted isomorphic blocks retain the normal-function problem. Absolute Hodge rigidity supplies Jacobian correspondences, but it does not by itself identify the relative extension classes. Condition (E5) is therefore load-bearing.

Finally, genuinely complex-valued phases on real two-chains have two-dimensional phase support. The real-line phase distribution and its one-variable Fourier–Laplace completion do not apply. No theorem for that category is claimed here.

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